F

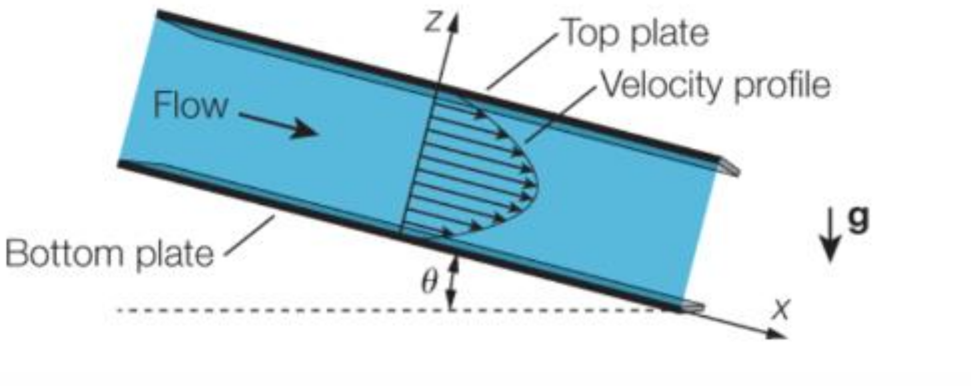
$x \quad d \quad \frac{s \quad \tau j}{} \frac{d}{} \quad F \; d$

$F \quad \frac{s \quad \tau j}{} \frac{d}{}$

$F \quad F$

$x \quad \frac{}{} \frac{s \quad \tau}{} \quad \frac{}{} \frac{s \quad \tau}{} \frac{d}{}$

$x \quad \frac{}{} \frac{s \quad \tau}{} \quad d$



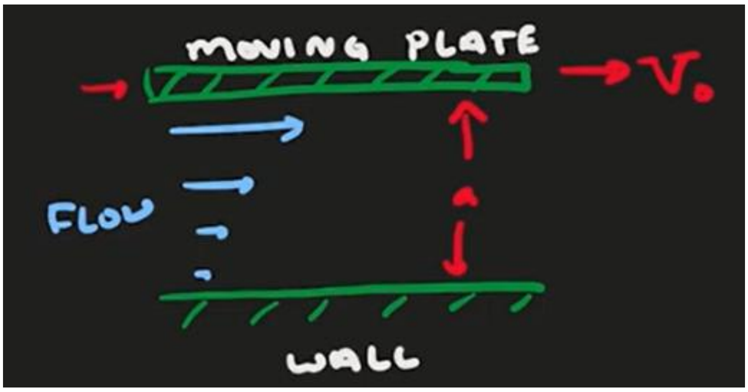
Escoamento plenamente desenvolvido dominado pelas forças friccionais

V s

s s s

d

d



τ vw

$\frac{-}{w}$

$\frac{x}{}$

$\frac{-}{}$ z

x 3 x d 3

Cz 3 z d

y 3 y d

$\frac{x}{}$ $C\frac{y}{}$ $\frac{z}{}$

$\frac{x}{}$ $\frac{z}{}$

$\frac{y}{}$

$\frac{y}{}g$ y vw F

y $_{3.d}$

F y

τ $\frac{x}{w}$ $x\frac{x}{}$ $y\frac{x}{}$ $z\frac{x}{}$ $\frac{s}{}$ $\frac{x}{}$ $\frac{x}{}$ $\frac{x}{}$ τ C

$\frac{x}{w}$

$x\frac{x}{}$ $x\frac{x}{}$

$z\frac{x}{}$ $\frac{x}{}$

τ

y $y\frac{x}{}$

$\frac{x}{}$

$\frac{x}{}$ F

x

x F F

x $_3$ x $_d$ $_3$

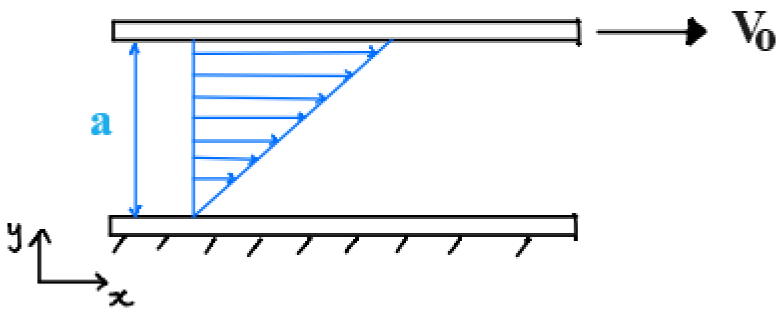
x $_3$ F F

F

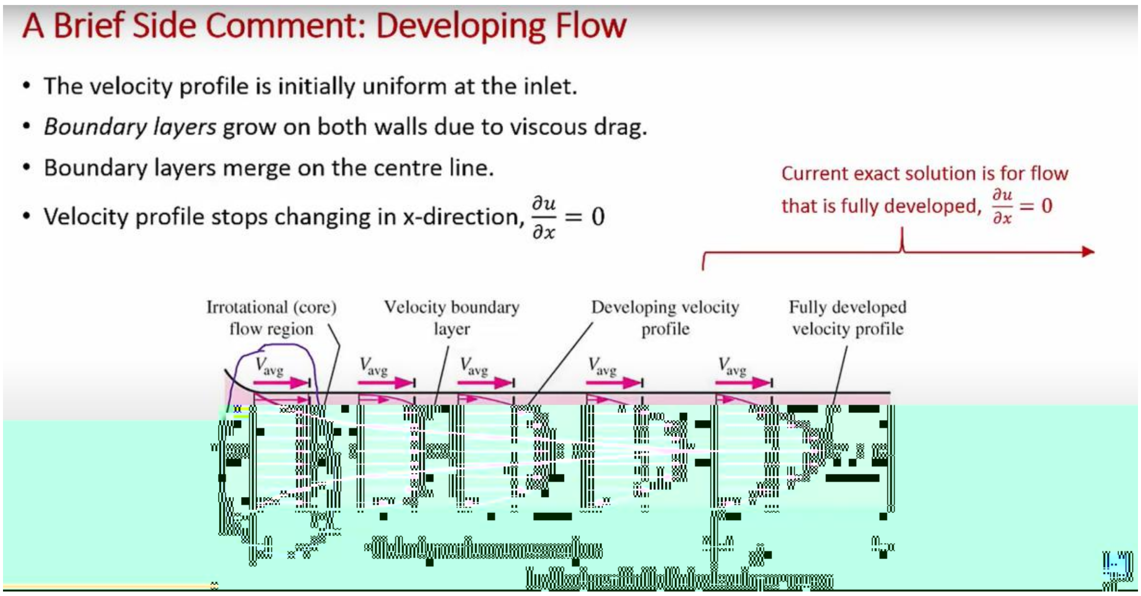
x $_d$ F d $_3$

F $\frac{}{d}$

x $\frac{3}{d}$ d



V0
a
y
x



A Brief Side Comment: Developing Flow
• The velocity profile is initially uniform at the inlet.
• Boundary layers grow on both walls due to viscous drag.
• Boundary layers merge on the centre line.
• Velocity profile stops changing in x-direction, $\frac{\partial u}{\partial x} = 0$
Current exact solution is for flow that is fully developed, $\frac{\partial u}{\partial x} = 0$
Irrotational (core) flow region
Velocity boundary layer
Developing velocity profile
Fully developed velocity profile
V_{avg}